

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	Medium

Question

	Human Resources	Accounting
Bachelor’s degree	4	3
Master’s degree	2	6

The table above shows the number of people who work in the Human Resources and Accounting departments of a company and the highest level of education they have completed. A person from one of these departments is to be chosen at random. If the person chosen works in the Human Resources department, what is the probability that the highest level of education the person completed is a master’s degree?

Answer

- A. $\frac{2}{15}$
- B. $\frac{1}{3}$
- C. $\frac{1}{4}$
- D. $\frac{8}{15}$

Correct Answer: B

Rationale

Choice B is correct. In total, there are 6 people in the Human Resources department. Of those 6, 2 have a master’s degree as their highest level of education. Therefore, the probability of an employee selected at random from the Human Resources department having a master’s degree is $\frac{2}{6}$, which simplifies to $\frac{1}{3}$.

Choice A is incorrect; it is the probability that an employee selected at random from either department will be in the Human Resources department and have a master’s degree. Choice C is incorrect; it is the probability that an employee with a master’s degree selected at random will be in the Human Resources department. Choice D is incorrect; it is the probability that an employee selected at random from either department will have a master’s degree.